

CHAPTER 1

1.0 INTRODUCTION

1.1 Background to Study

Storage is one of the most essential parts of the agricultural process. This is due to the propensity of food materials to spoil, there causing loss of value in the food material in question. Humans always required food to consume. However, for various reasons, it was not always obtainable (Hammond et al. 2015). These reasons include the seasonality of food plants, occurrence of drought and overpopulation. Man adapts to this situation by storing food during times of excess.

Storage is as old as man himself. Early human hunter-gathering groups are known to have practiced basic methods of storage. There is evidence for the use of smoking and sealing being used to store food (Burger and Fristoe, 2018). An innovation worth mentioning here is the observation of stored food under ice sheets in the cold regions of the north pole. This may represent the first time humans came into contact with the preservative properties of coldness. Modern technological advancements have giving rise to more efficient and energy intensive storage methods including freezing, pasteurization, sterilization and so on which each method catering to a subset of food materials.

Numbering among recent innovations in the storage of food materials is modified atmospheric storage. This involves the provision of an appropriate protective atmosphere around the food material being stored (Majidi et al, 2014). The atmosphere around such product reduce microbial growth, respiration and enzymatic reactions which most food materials experience as a result of the presence of oxygen. The operating principle behind this was known since the age of science. It was then applied to the storage of meats during the 1930s (Inns, 1987). Ever since then, it has been applied to an ever-growing list of perishable food items.

Modified atmospheric storage methods is still a developing field of study with improvements being made on the gas composition required, the energy requirements and length of shelf life preservation for specific food products.

1.2 Statement of the Problem

Not much attention has been given to improvement of storage methods compared to the increase of agricultural production capacity in terms of research and funding for research. This is due to a variety of reasons, such as unavailability of accurate measurements of post-harvest losses, the subtlety of the losses involved or the assumption by a majority of the public that such problems have been satisfactorily solved by the scientific community. Indeed, man has been able to increase the shelf life of food materials to long durations and such technologies are not uncommon. However, the estimates concerning food losses in general do not back up this assumption. Between 702 and 828 million people faced hunger in 2021 (FAO *et al.*, 2021). Out of that, 278 million were in Africa. This is contrasted the amount of food wasted. According to world food counts, approximately 1.3 billion tons of human consumable food is wasted every year. That's approximately one-third of the estimated total food production of the world. The same source also stated that post-harvest handling and storage wastage contributed the second largest number, led by production operations. In Africa, the numbers is much more. Estimated losses range between 30 and 50 percent of production. In sub-Saharan, fruit and vegetable losses are up to 52 percent with post-harvest handling and storage contributing to 8 percent or more of those losses.

From the numbers given above, it is evident that the world's food scarcity problem is not entirely one of inadequate production. Food security, in many cases, is a question of access rather than availability. In the cases of most perishable goods, it is simply uneconomical to transmit them to the market for sell as they do not have adequate shelf life. Modified

atmospheric storage techniques provide a solution to these problems but, like most high performing storage methods, have high energy and high-cost demands making it inaccessible to most farmers requiring them. Innovation and improvisation are therefore needed to reduce these costs and make such techniques accessible.

1.3 Objective of Study

The overall aim of this project is to evaluate the viability of an improvised modified atmospheric storage technique on the shelf life of a perishable farm produce. Specific objectives of the study include:

1. To use dry ice to produce oxygen deficient environments in an enclosed container.
2. To determine if those oxygen deficient atmospheres can cause an increase in the shelf life of the perishable farm produce.

1.4 Justification

Storage of perishable goods is necessary to prevent food scarcity and ensure food security for any population. It is then essential to ensure research in food storage and preservation methods continues to improve the shortcomings of existing food storage systems and to innovate to provide solution to specific problems within the storage systems. This study will research into modified atmosphere storage methods to improvise it for areas and people lacking access modern storage infrastructure.

1.5 Scope of Study

This research is limited to studying tomato as a perishable product. Removal of oxygen via the use of dry ice sublimation produced modified atmospheres. The parameters measured to quantify spoilage includes PH value, total alkalinity, total solids and titratable acid.

CHAPTER 2

2.0 LITERATURE REVIEW

2.1 Fruit and vegetable: a brief overview

Fruits and vegetables are plant-based foods that are an essential part of a healthy and balanced diet. They are rich in essential vitamins, minerals, fibre, and other nutrients that are important for good health. However, the botanical definition of fruits and vegetables can be more specific. Fruits are the mature ovary of a flowering plant that contains the seeds of the plant. They can be classified as either fleshy or dry fruits. Fleshy fruits have a fleshy edible portion that surrounds the seeds, while dry fruits have a hard outer shell that protects the seeds. Examples of fleshy fruits include apples, oranges, and berries, while examples of dry fruits include nuts and grains. Vegetables, on the other hand, are derived from various plant parts such as leaves, stems, and roots. They can be classified into several categories based on their botanical origin: leafy vegetables such as lettuce and spinach, stem vegetables such as asparagus and celery, root vegetables such as carrots and potatoes, and flower vegetables such as broccoli and cauliflower.

2.2 Tomato as a case study

A tomato fruit is spherical to oval shaped fruit which is flesh in nature and contains at least two cells of small seeds in each sample. It is usually coloured red, yellow or scarlet although purple and green varieties exist. Its scientific name is *Solanum Lycopersicum*. There have been lots of arguments concerning whether it is a fruit or a vegetable. Botanically, it is both a fruit and a vegetable. Fruits are botanically defined as a ripened flower ovary that contains seed. In the case of tomato, this is true. There is a lot of vagueness concerning the definition of a vegetable. However, the World Vegetable Centre proposed to define vegetables as “most herbaceous annual plants of which some portion is eaten, either cooked or raw, during the principal part of the meal to complement starchy food and other food items”. Tomato therefore also falls in this category. The average shelf life for fresh tomato is one week. Refrigeration is known to increase it to two weeks. Harvesting it at the right time is also known to increase its shelf life as puncture

of the skin becomes less probable (Ullah, 2009). Tomato has many health benefits for humans. It helps regulate blood pressure, prevents eye problems, and helps maintain a healthy skin.

2.3 Fruit and vegetable production and uses.

Fruit and vegetable production is a vital aspect of agriculture and plays a significant role in human nutrition. Fruits and vegetables are an important source of essential nutrients such as vitamins, minerals, fibre, and antioxidants, which help to maintain a healthy body and prevent chronic diseases. The cultivation of fruits and vegetables involves various processes, such as soil preparation, planting, fertilization, irrigation, pest and disease management, and harvesting. Different fruits and vegetables have unique growth requirements, and it is important to consider factors such as soil type, climate, and water availability to achieve optimal production. Once harvested, fruits and vegetables have a wide range of uses, both culinary and non-culinary. In the culinary world, fruits and vegetables are used in a variety of ways, such as raw consumption, cooking, baking, and juicing. They can be used to make salads, soups, stews, sauces, smoothies, and desserts, among other things. Fruits and vegetables are also used in the production of various non-culinary products such as cosmetics, medicines, and biofuels. For example, there are studies which look into the use of fruits and vegetables as solutions to cancer and cardiovascular diseases (van't Veer, 2000).

In addition to their nutritional and economic value, fruit and vegetable production also plays a critical role in environmental sustainability. Fruits and vegetables are often grown using sustainable agricultural practices such as organic farming, which helps to reduce the use of synthetic fertilizers and pesticides and promotes soil health and biodiversity.

The estimated annual production levels for some of these fruits and vegetables in Nigeria are very high. Certain estimates put onions at 3.8 million tonnes, tomatoes at 6 million tonnes, plantain at 15 million tonnes and so on (Ibeawuchi et al., 2015). Different fruits and vegetables

are farmed predominantly in different parts of the country because of difference in climates. However, the number of large-scale fruit and vegetable processing industries in Nigeria is low, and the small and medium-scale plants do not match the country's potential. This means that most of uses of fruits and vegetables a consumed in its natural state, without much processing greatly reducing the potential profit of the agricultural endeavour. The main cause of this lies in Nigeria's reliance on subsistence farming, leaving no room for industrial expansion.

2.4 Food spoilage and losses

In essence, food spoilage is the process where a food product becomes unsuitable to be ingested by the consumer (Maru, 2022). It may be defined as the process leading to a product becoming either undesirable or unacceptable for human consumption with an associated change in smell, taste, and colour.

Further reflection about food spoilage will reveal it to be part of a natural process known as decomposition. Decomposition is defined as “a biological process that includes the physical breakdown and biochemical transformation of complex organic molecules of dead material into simpler organic and inorganic molecules” (Juma, 1998). As a result of the definitions stated above, it can be inferred that food spoilage is the form decomposition affecting organic products that man ascribes nutritional or economic value. This is the view that would be taken by this project.

In real world situations, food is considered as spoiled if it develops certain characteristics that are not desired by customers even though such characteristics are not harmful to humans. For example, Potato samples are usually discarded instead of being processed due to Net necrosis. Net necrosis are tiny areas of dead cells found in potato tubers. Infected potato samples are not particularly harmful but are undesirable by customers and are thereby considered (in certain commercial processes) as spoiled. As seen in the example given

above, deteriorating foods may be safe to eat, having no poisonous effects for humans and animals. However, changes in texture, smell and appearance may cause potential consumers to reject it (Doyle, 2007). In this project, such cases would also be considered as spoilage.

Food deterioration is unavoidable. This is because, by all indications, crop produces begin to decompose immediately after its harvest as crops loses their biological resistance to spoilage at the harvesting point (Dauthy, 1995). The only option remaining for an individual is to preserve the quality of the food produce to the point of consumption without it getting spoilt. This requires a basic understanding of the types of spoilage that may occur and how to prevent them. The types of food spoilage include:

- Microbial Spoilage.
- Physical Spoilage.
- Chemical spoilage.

2.4.1 Microbial Spoilage: Refers to spoilage caused by the action of micro-organisms. This is seen as the main cause of spoilage and deterioration in general. The word “micro” is defined by the Merriam Webster dictionary to mean very small. An organism is defined by Mosby’s dictionary of medicine, nursing, and health profession to mean any living organic system which functions as an individual entity. Therefore, micro-organisms can be defined as independent organic systems which exhibit all the characteristics of living things and are too small to be seen by the naked eye. The size of the micro-organisms or microbes varies widely according to the type in question. Bacteria, a type of microbe, has an average diameter of 1 to 2 microbes. There are also microbes with diameters of ten micrometre. Generally, microbes require microscopes to be observed.

There are many types of micro-organisms that cause spoilage, and knowledge about these causes has aided man to prevent loss due to spoilage and contamination. Many studies

have been carried out with the aim of understanding specific species and how they affect food substances. However, it would be more useful to generalize these spoilage agents into macro-groups (or families) that share the same characteristics if economic effectiveness is a goal in the storage process. This then enables us to effectively study them and deduce preventive methods for them. The main microbic causes, or agents of spoilage are (Dauthy, 1995):

- Bacteria
- Yeasts
- Moulds

Bacteria (Bacterium in singular) are microbic organisms belonging to the kingdom Monera and therefore have no nucleus and no mitochondria (Jones and Jones, 2014). They are often single celled and have cell walls that are not made of cellulose. They carry out life's functions individually and are called unicellular. Bacteria can be classified into five groups based on their shape; cocci (spherical shaped), bacilli (rod shaped), spirilla (spiral shaped), vibrios (comma shaped) and spirochaetes (corkscrew shaped). Most bacteria are not particularly harmful to humans and animals. Instead, many of them are useful for bio-chemical processes within other large organisms (such as bacteria in human's digestive tract), and some help in the cycles of various elements i.e., nitrogen and carbon. They also help in the decomposition of organic matter. Although the advantages of bacteria are quite much, they also cause harm to the environment and other organisms. They do so in three ways; producing harmful by-products (i.e., the chemical botulin by clostridium), cause diseases in plants and animals and produce unwanted decomposition of organic materials otherwise known as spoilage (Lundsby, 1984). Bacteria responsible for spoilage includes: *Brochothrix* spp, *Carnobacterium* spp, *Lactobacillus* spp, and so on.

Bacteria is found everywhere, from the air to water. They are therefore considered biologically ubiquitous. They are everywhere because of their mode of reproduction; binary fission. Binary fission is an essential factor in the bacteria's success in the ecological system of plant earth. The process of binary fission involves a bacteria cell growing to twice its size and splitting itself after reproducing double of every living part of itself. The product of a successful binary fission is two viable bacteria cells. The rate at which this reproduction method is carried out in bacteria is mind boggling. In favorable conditions, one cell may complete this process in just 20 minutes. That translates to a colony of just above 1 million cells in approximately seven hours! As a result of this, it is very hard to prevent contamination of food produce by bacteria. They are very viable and reproduce faster than most micro-organisms.

When conditions are unfavorable for growth, certain types of bacteria can produce spore-like structures and become Endospores. These enables the bacteria to retail its basic life functioning information while surviving in harsh environmental conditions. These Endospores are bacteria covered in a hard shell and can survive for years if necessary. Examples of spore producing bacteria includes the *Bacillus* and *Clostridium* species. These types of bacteria are particularly threatening for heat treated foods as they don't typically die from heat treatment. They are also resistant to freezing temperatures and grow on food stored in refrigerators and cold rooms, especially meats (Lundsby, 1984).

Conditions necessary for bacteria growth vary widely. Some bacteria require oxygen (aerobic bacteria i.e., *Escherichia coli*), some do not (anaerobic bacteria i.e., *Lactobacillus*) and still others are harmed by oxygen (obligate anaerobes i.e., *Clostridium botulinum*). Others thrive in cold environments (*Psychrophiles* i.e., *Antrobacter globiformis*) others require warm surroundings (*Mesophiles* i.e., *Legionella*) and still others require hot environments (*thermophiles* i.e., *Alicyclobacillus acidoterrestris*). However, we may generalize by stating

that bacteria do better in a warm and protein rich environment with moisture and a neutral PH level.

Yeasts: Yeasts are single celled organisms in the kingdom Fungi that have a nucleus membrane. They do not have a population and specie diversity that matches bacteria but can still be found everywhere including in the soil freshwater, seawater and the air. Yeasts are not known to be particularly pathogenic to animals. Most yeast cells have diameter normal for most micro-organism except for “giant yeast”, yeast cells of the size 40 micrometers that have been observed (Wu *et al*, 2002). Some species, such as *Saccharomyces cerevisiae*, is used to extensively to leaven bread in baking. *Saccharomyces uvarum*, another specie of yeast used industrially to ferment alcohols with *Saccharomyces cerevisiae* as a viable alternative. However, yeasts are still spoilage agents causing economic loses for farmers. Types of yeast that cause spoilage includes *Zygosaccharomyces*, *Candida* and *Saccharomyces*.

Yeasts have two modes of reproduction: mitosis and budding. Most yeasts use mitosis. Mitosis involves the splitting of a yeast cell into two cells having the same chromosome number and kind. This involves the elongation of the nucleus, the division and splitting of the nucleus and the splitting of the cell membrane with one nucleus with a surrounding cytoplasm. This is like binary fission. However binary fission occurs much faster and only in organisms with no nucleus membrane (Prokaryotes). Budding on the other hand is the asymmetrical splitting of the yeast cytoplasm after the division of the nucleus has taken place. Yeast may also produce spores but only under stress.

The general conditions necessary for yeast growth vary widely with respect to the species in question. However, generalizations can be made. Yeasts do not thrive in hot environments, the temperature range been between several degrees below 0oC and several degrees above 50oC . Pressure not particularly a factor for yeast cells on land although some of its species withstand

large hydrostatic pressures under the sea. Also, yeasts require a slightly acidic medium to thrive with values ranging between 4.5 and 5.5 on the PH scale. This is not a rule though due to yeasts exceptional ability to withstand large variations in PH values. Yeasts are more tolerant of acidic levels with a general tolerance range of 3 to 8 on the PH scale.

Moulds: Moulds are multicellular filamentous organisms under the kingdom Fungi in biological classification. Sometimes called Mildew, they represent a collection of micro-organisms that can be observed with the naked eye as a white covering or mass over damp or decaying substances. The mould growth on such substances are basically collections of filaments called hydra. Each Hydra intertwine with others to covering the infected substance and forming a mycelium. Mycelium's usually looks white but black and Smokey are not unheard of. Coloured moulds can be observed in matured mould colonies. Moulds are used extensively in the production of certain cheese and in the manufacturing of chemicals. One of such chemicals is penicillin, an effective antibiotic produce by moulds from the genus penicillium. They also happen to infect food substances such as bread and cheese causing soilage.

Moulds reproduce using spore formation. The spores released by the moulds are light weight and buoyant in air giving them high access range. After landing, they may begin to germinate. Spores of mould that landed on surfaces do require moisture to germinate. Their requirement for water, however, is not as necessary as yeast. This makes it able to survive in almost any surface. They even grow on cloths and on the walls of houses. This strength in survival is not equally represented by its spores as they are not heat resistant in comparison to other specie spores i.e., bacteria.

The general rule of in nature is the survival of the fittest. Any organism that has the ability takes the nutrient for itself leaving others to starve. Combining this statement with the fact that most

micro-organisms can be found anywhere, and we may conclude that microbes of different types may end up competing against each other for food resources. The microbiological profile of a food material before it is stored and after is usually very different. Before spoilage occurs, varied counts of microbe types exist with an associated population per each type. This includes bacteria, fungi, and even certain viruses. After spoilage only one or two become the predominant population in the food material (Lorenzo et. al., 2018). The identity of these microbes depends on the type of food to be consumed, the speed of reproduction of individual microbe types and the environmental conditions in which the food material is placed in. Bacteria tends to be the most viable because of its reproductive mechanism, binary fission. However, certain conditions may prevent it from growing effectively. It is quite known that bacteria and yeasts require more moisture than moulds to survive. For this reason, moulds are seen to grow better in semi-dry foods (Dauthy, 1995). Also, Yeasts are known to grow better in low PH values and low moisture mediums than most bacteria enabling it to form colonies acidic crops such as grapes (Bullerman, 2003). However, these are just exceptions. In most cases, certain forms of bacteria take over stored food materials causing spoilage.

2.4.2 Chemical Spoilage: in the case refers to an unwanted change in appearance, colour, smell, or taste of a produce due to the action of chemicals on the produce. Such chemicals could be enzymes, oxidizing or reducing in nature. They are, however, not living. A large portion of these chemicals are produced by enzymes. Enzymes are substances found in living organisms that speed up specific bio-chemical reactions without changing its chemical composition. They are almost always proteins. Enzymes cause a variety of changes in foods where it is present. Some of these changes are deemed useful i.e., the ripening of banana. However, they are also a major contributor to deterioration by catalysing unwanted chemical reactions resulting in unwanted and sometimes poisonous chemicals.

Enzymes are affected by environmental factors that determine its effectiveness in catalysing biochemical reactions. Such factors include:

1. Temperature.
2. Available water.
3. Acidity or alkalinity.
4. Presence of enhancing or inhibiting chemicals.

The usual temperature range for enzymes is 0oC to 60oC with 37oC being the optimum temperature. Enzymes are deactivated at temperatures that are too hot, about 80oC and are slowed down by low temperatures.

Micro-organisms also produce harmful by-products. An example is mould. Moulds are responsible for producing poisonous chemicals called Mycotoxins. Mycotoxins are known to cause illness and, in extreme cases, death to animals that consume foods filled with them.

2.4.3 Physical Spoilage: refers to spoilage caused by physical damage to food during harvesting, processing, or distribution. During physical spoilage, the protective outer layer of the crop is opened or more accurately, broken. This leaves the produce open to influx of harmful micro-organism and chemicals. Physical spoilage starts during harvest as many farmers are ignorant to proper handling techniques. Coupled with improper storing such as heaping tomatoes thereby compressing the bottom fruits, microbes and chemicals can more easily access the food material and begin. Spoilage process. Mastery of proper handling techniques greatly increase the storage time of fruits and vegetables.

2.5 Food storage

According to Britannica encyclopaedia, food are substances containing carbohydrates, proteins, fats and other nutrients needed by the body of organisms to sustain growth and essential processes and to provide energy. The implication of the above definition is not to hard

to see. Not all substances considered as food for one organism can be considered as food for the other. Some substances may even be poisonous for one organism while being necessary for another. In this project, food would refer to substances that provide essential nutrients to humans only, except it is specified otherwise.

Storage refers to the phase of a post-harvest system during which food materials are kept in such a way as to guarantee food security (Lucia and Assennato, 1994). It may also be defined as the process of extending the natural shelf life of agricultural products for future use as food, raw materials, fuel or for sale. It may be closely related with preservation, which is a process of avoiding deterioration or spoilage of agricultural produce by micro-organisms to ensure its adequate supply.

2.5.1 History of storage methods: Ancient cultures are known to have practiced drying, smoking and salting (Hammond et al., 2015). They stored berries and nuts during wanderings. There is even evidence of them digging underground storage pits. The Sannai Maruyama site in Aomori, Japan contains storage pits as deep as 2 meters used by hunter-gathers to store foods believed to be chestnuts for future use. There is also evidence of cultures in the American continent doing the same thing (Habu, 2008). These methods of storage were prevalent for most of man's stay on earth. They are still used extensively today.

However, these were not the only methods used to store food. The ancient Greeks and Romans were known to store fruits in honey. Honey wasn't always available though, so jam and jelly was used extensively (Nummer and Brian, 2002). Merchants carrying their goods in ships were particularly fond of this method. Fermentation was also used, unknowing taking advantage of microbes to convert starch derived sugars into alcohol (Nummer and Brian, 2002). The beauty of this method was not just the preservation of the food material but the production of a more palatable and nutritious end-product. Food items used to produce alcohols include barley,

wheat, rice and so on. Pickling was also discovered, which involves the storage of foods in vinegar or other acids. This method was popular during the sixteenth century in Europe (Nummer and Brian, 2002).

As understanding of the causes of spoilage increased, more sophisticated methods were developed to increase shelf life of desired products. One of such modern method is canning, first pioneered by Nicolas Appert, a French inventor in the 1790s who discovered that the application of heat to sealed glass bottles with food extended the food's shelf life. Although its processes were not understood, his work helped others in their quest to find viable storage methods culminating in canning's mass adoption during the 1920 (). Another method developed during man's transition to the modern age is pasteurization. Discovered by Louis Pasteur, this method helped extend the shelf life of many agricultural products such as milk. Much more recent discoveries in modern storage includes irradiation, high pressure food preservation and pulsed electric field (Amit et al, 2017).

The main purpose for storing food substances is to extend its shelf life while preventing significant decline in food quality (McCurdy, 2009). All other reasons are derivatives from this. The specific purpose for storage differs on account of the entity carrying it out. For example, a home manager may store food to ensure it may be eaten at a date that is past its ordinary shelf life while a farmer may be more concerned with preserving seeds for future planting. According to FAO, the main objectives of the storage of perishable goods includes the following:

1. To permit deferred use of the food materials obtained on a specified time period.
2. To ensure the availability of viable seeds for future planting seasons.
3. To ensure regular and adequate supply of raw materials to processing industries.
4. To stabilize the market prices of selected crops by balancing the forces of supply and demand.

Food spoilage is a major contributing factor to food scarcity in any society experiencing it. This is because, as stated in the introduction chapter, most food scarcity situations are more of an issue of the inability of food distribution systems to access areas lacking in food resources.

2.5.2 Food storage methods

The methods of storage of perishable goods are many and may be classified by many criteria. Some of these classifications includes:

1. In accordance with technological sophistication of storage method. Such classifications include traditional and modern storage methods.
2. In accordance to continuance or discontinuance of crop material. Such classifications include bulk and bag storage methods.
3. In accordance to the time period of intended storage i.e. duration of storage. Such classification includes short term, medium term and long-term storage methods.
4. In accordance to size or scale of storage. This includes small scale, medium scale, and large-scale storage methods.
5. In accordance to the principle of storage. This includes chemical, biological and physical storage methods.

In accordance with classification based on principle of storage, we have:

Chemical storage methods. These types of storage methods make use chemicals preservatives to prevent spoilage of agricultural materials. Chemicals used to prevent deterioration of food products includes phosphine dust, wax, or atelic dust. Chemical preservatives are substances added in small quantities which help prevent spoilage while doing little to alter the physio-chemical properties of the food materials. Herein lies a serious advantage of using chemical storage methods; they do not induce significant change in the physical form or chemical

structure of the product being stored. They help prevent respiration and discourage insect infestation. Chemical storage methods tend to be very effective and economical for the individuals who use them. However, many of these methods have been shown to be unsafe as their equivalent chemical preservatives are poisonous or react with the food to form poisonous substances. These statements were extracted from an FAO paper on the use of phosphine dust: “Because phosphine dust is highly toxic, inhalation of even small quantities of the dust from formulation, as well as the evolved should be avoided” and that exposure to unhealthy concentrations may cause “...symptoms consisting of fatigue, ringing of the ears, nausea, or pressure in the chest...” (Bond, 1984).

General rules for chemical preservation include the following:

1. Chemical preservatives should be used with levels that are necessary for normal preservation and not more.
2. Addition of new preservatives to stop microbe deterioration is not recommended.
3. Chemical preservatives used must be recognized as not having any harmful effect on human's health and accepted by national and international laws (Dauthy, 1995).

Biological storage methods. This classification of storage methods involves the use of biological agents to prevent the deterioration of agricultural produce. This is very much associated with Bio-preservation. Bio-preservation can be seen as the “the extension of shelf life achieved by the use of a natural or controlled microbiota and/or their antimicrobial compounds” (Stiles, 1996). The main example of a biological storage method is fermentation. Fermentation can be defined as the decomposition of carbohydrates i.e., sugars and starches by the action of micro-organisms and /or enzymes (Amit et al, 2017). The product of fermentation is very nutritious and tasty to humans. Fermentation is carried out on a wide variety of agricultural products such as wheat, barley, rice and so on by an even wider variety of microbes.

Physical storage methods. This classification of storage methods operates using physical principles to preserve the quality of stored agricultural materials. It involves the manipulation of physical quantities with the aim of extending the shelf life of the food being stored. This is done by monitoring the physical environment around the agricultural produces and ensuring a longer shelf life by controlling the values of specified physical parameters.

Most of the activities contributing to the deterioration of farm produces i.e., microbial activities and enzyme induced reactions are mainly affected by temperature (Majidi et al, 2014). As a result, temperature can be seen as a primary spoilage factor. This is indicated by the fact that most physical storage methods alter the temperature of the produce's surroundings in some way. Therefore, Physical storage may be broadly divided into thermal and non-thermal storage methods (Amit, 2017). Thermal storage methods are those storage methods which functions by the addition or removal of heat energy with respect to the food material being stored. Such methods include:

1. Freeze storage: Freezing is the process of lowering the temperature of a food material to temperatures below -18 degrees Celsius or lower thereby preventing microbial growth and enzymatic reactions. Almost any type of perishable food material can be stored by freezing. It is especially popular with meat and animal product storage because the length at which shelf life is extended. Perishable products are known to last up to 20 days under such a process. Freeze storage appear in industrial processes in the form of warehouses which maintain certain temperatures artificially, or cold rooms. Other applications of freeze storage are freezers, used for storage commercially and domestically.
2. Pasteurization: This is often seen as a mild heat treatment process as it is carried out at temperature values below zero. Pasteurization is the heat treatment of food materials with the purpose of stopping enzymatic change and microbial growth with minimal

change in its form and nutrient content. It is done when change of the form or sensory properties of the food is not desired. Its preservation duration differs with the product being stored. Milk may be preserved for a couple of days while fruit jams and alcohols may be sealed and pasteurized to last months. Types of pasteurization includes thermization, batch pasteurization and flash pasteurization.

3. **Canning:** This is a thermal process involving the sealing of food materials in air-tight containers before, during or after sterilization. The containers are usually made up of tin although plastics and glass are used sometimes. Many of these containers are also coated to avoid corrosion from acidic contents. Sterilization is the complete destruction of all viable organisms in a selected material. In industrial operations, this process destroys all nutrients. Therefore, only commercial sterilization is desired. Commercial sterilization is the process whereby a product is optimally processed to prevent deterioration even in normal conditions. In canning, this involves heating a material to a temperature between 110 and 125. Types of canning includes water bath canning, pressure canning and atmosphere steam canning.

Other methods of storage include drying and refrigeration.

Non-thermal storage methods are those storage methods that do not rely on the transfer of heat to prevent spoilage. Such methods include:

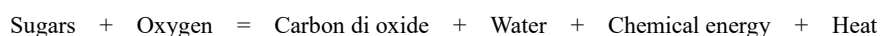
1. **Irradiation:** This is the deliberate exposure of food material to radiation to extend shelf life (Yousefi and Razdari, 2014). It involves the exposure of food materials to ionizing radiation such as gamma rays, electron beams and x-rays (Farkas, 2006). The most used form is the gamma ray type because it is produced naturally in cobalt 60 and cesium 137. Irradiations advantage lies in the fact that they prevent microbes from reproducing without causing substantial change in the properties of the food. This is because

irradiation ray break the DNA strands of microbes quite easily, making it impossible for them to reproduce effectively (Yousefi and Razdari, 2014). Irradiation is becoming a widely used storage method in our modern world.

2. High hydrostatic pressure: Is a method of non-thermal storage that affects enzymes and micro-organisms by ultra-high-pressure values. Such values could reach 600MPa in some cases (Buzrul, 2015). The DNA strands of the enzymes and micro-organisms are then put under heavy stress, inactivating them (Xia et al, 2022). The result is a well-preserved food material with little change in its nutrient and sensory properties. The pressure values of ultra-high-pressure methods are usually gotten by water, thereby the name high hydrostatic pressure storage.

Controlled and modified atmosphere storage: These methods work by altering the composition of the ambient air around the food material being stored to obtain an optimized shelf life. The gas composition of the air around a produce is usually considered secondary to temperature when storing materials. Nevertheless, its manipulation is a powerful tool in controlling the quality of fresh produce (Ben-Yehoshua et al, 2005). Controlled and modified atmosphere storage methods prevent spoilage in produces by countering three modes in which spoilage occurs in produce, namely microbial action, respiration and enzymatic ripening.

- Microbial activities are curtailed by the absence of adequate levels of oxygen. Addition of carbon di oxide is also known to suffocate many micro-organisms, even anaerobic types.
- Respiration occurs in crops by the following general reaction:



By removing oxygen from the atmosphere around the produce, this transformation is halted preventing the release of energy which in normal circumstances deteriorates the produce (Ben-Yehoshua et al, 2005).

- Enzymatic ripening is known to be caused by the action of ethylene on mainly climatic fruits (Falagán and Terry, 2018). Increase in Carbon di oxide levels and decrease in oxygen levels have been proven to inhibit the production ethylene gas from fruits, thereby slowing their ripening (Ben-Yehoshua et al, 2005). Furthermore, climacteric fruits are known to increase respiration during ripening. Therefore, the postponement of ripening prevents excess respiration and further extends the products shelf life.

Controlled and modified atmospheric storage do not mean the same thing. Controlled atmospheric storage deals with the strict monitoring and maintenance of predetermined optimal gas proportions in the ambient air surrounding the food material while modified atmosphere storage involves packaging food materials in an airtight container with an optimized gas proportion without monitoring, thereby allowing the changing of gas composition as the material interacts with the air. Controlled atmosphere storage methods have high energy requirements compared to modified types. Due to this reason, controlled atmosphere storage methods are only used for long term storage of products such as pears and apples (Ben-Yehoshua et al, 2005).

CHAPTER 3

3.0 MATERIALS AND METHODS

3.1 Materials

The material used in this project are tomato samples gotten from the local area (kure market, Minna, Niger state).

3.1.1 Apparatus

Apparatus proposed to be used in this project would include a burette, fourteen airtight containers, the desiccator, measuring cylinder, a pipette, a crucible, an electric weighing balance, a conical flask, a petri dish, a CO₂ container, a tube rack, a desiccator, selenium catalyst, a spatula, a cheese cloth, filter paper, wool gloves, an oven dryer, and a refractometer.

3.1.2 Reagents

Reagents used in this project include sulfuric acid (concentrated), hydrochloric acid, methyl red indicator, Phenolphthalein indicator, and sodium hydroxide solution.

3.2 Methods

3.2.1 Sample collection and preparation

Sample preparation was done using the following procedure:

Fresh samples of the perishable food material, tomato, was bought from kure market in Minna, Niger state. The perishable food was then washed gently with water to avoid brushing and puncturing of the material. No soap was added to the water to avoid possible unwanted chemical reactions. The perishable samples were then dried with a soft wool cloth. To select samples for storage, no attention was given to sample size. Instead, the samples were graded to obtain samples with the least spoilage signs. Spoilage signs guarded against include uncharacteristic softness and smell. Also, samples with bruises, cuts and black moulds that

extended past skin level were rejected. The selected samples were then set aside and used to experimentation.

3.2.2 Container preparation.

Fourteen airtight containers were obtained. The containers were made of glass and plastic to ensure inertness. All containers were washed thoroughly with clean water only so as not to introduce unknown chemical residues. They were then divided into two groups with seven containers each. One group was specified for the atmospheric storage method while the other group was specified as control.

3.2.3 Improvised atmospheric modification used.

Atmosphere modification using dry ice was carried out using the following procedure:

The first group of containers were given tomato samples and sealed shut. Dry ice was then obtained from the CO₂ container by deposition on a clean cloth. The dry ice was taken using wool gloves, placed in each of the second group of containers and allowed to sublime completely. Sublimation occurred until the gas was observed to exit the containers. Tomato samples were then gently placed in each container and sealed shut. Each day, two samples, one from each group, were taken and analyzed for the previously stated parameters. The experiment ran for seven days.

3.2.4 Parameters determined.

Parameters determined include the following: PH value, titratable acid, total solids, and total alkalinity.

3.2.5 Procedure for obtaining parameters.

At daily interval one sample from each from the control and stored group was used to obtain two extract samples. Each representative sample was crushed with a ceramic mortar and

pestle thoroughly. The resultant product was then filtered by a fine filtering cloth to obtain the tomato extract used in determining the various parameters.

3.2.5.1 Total alkalinity determination

A representative sample of the solution was obtained using a clean and dry container and a few drops of phenolphthalein indicator were added to the sample. Using a clean, dry container, a sample of the solution that was indicative of the whole was taken. Standardized hydrochloric acid (HCl) solution was put into a burette and titration was carried out. After all of the alkalinities had been neutralized the volume of hydrochloric acid used was recorded. The following formula was then used:

$$\text{Total Alkalinity factor} = (\text{Volume of HCl used}) / (\text{Volume of extract used})$$

3.2.5.2 Total solids determination

A representative sample of the solution was obtained using a clean and dry container. The sample was then weighed using a tared container and the weight was recorded. The sample was placed in a pre-weighed evaporating dish and heated on a hot plate or in an oven at a temperature of 103-105 °C until a constant weight was obtained. The evaporating dish with the residue was then cooled to room temperature in a desiccator and weighed. The weight of the evaporating dish with the residue was subtracted from the weight of the evaporating dish and sample to obtain the weight of the solids. The total solids were calculated as a percentage of the original sample weight using the formula:

$$\text{Total Solids (\%)} = (\text{Weight of solids} / \text{Weight of sample}) \times 100.$$

3.2.5.3 Titratable acid determination

A sample of the solution was obtained using a clean and dry container. A precise volume of the sample was measured using a volumetric pipette and transferred into a clean Erlenmeyer flask.

A few drops of phenolphthalein indicator were added to the flask. Standardized sodium hydroxide (NaOH) solution was slowly titrated into the sample while stirring until a pink color persisted for at least 30 seconds. The volume of NaOH used for titration was recorded. The titratable acidity was then calculated using the following formula:

Titrateable Acidity (TA) = (Volume of NaOH used x Normality of NaOH x 100) / Mass of sample

3.2.5.4 PH value determination

The PH meter was first calibrated according to the manufacturer's instructions. Then, the electrode was rinsed with distilled water and blotted dry. A small amount of the extract was added to a clean container and mixed with distilled water to make a solution if necessary. Next, the pH meter electrode was immersed into the solution, ensuring it was fully submerged. The reading was allowed to stabilize, and the pH value displayed on the meter was recorded. Finally, the electrode was cleaned with distilled water and blotted dry before being used for subsequent measurements.

CHAPTER 4

4.0 RESULTS AND DISCUSSION

4.1 Results

The results of the storage of tomato samples in air and using CO₂ are represented in the tables below.

Table 4.1 Showing the variation of stated parameters in the control samples in relation to time in days.

Control measurements				
Day	PH	Titrateable acidity (g/kg)	Total alkalinity factor	Total solids (%)
0	4.18	2.8233	1.5	5.3
1	4.2	3.0267	2	5.1
2	4.29	1.2633	0.8	6.9
3	4.85	0.6409	0.13	3.4
4	4.24	6.2385	0.09	1.7
5	4.04	4.6005	0.05	1.8

Table 4.2 Showing the variation of stated parameters in the improvised modified atmospheric storage samples in relation to time in days.

Stored measurements				
Day	PH	Titrateable acidity (g/kg)	Total alkalinity factor	Total solids (%)
0	4.18	2.8233	1.5	5.3
1	3.91	2.5368	0.8	3.9
2	4.8	2.4017	0.6	3.2

3	4.6	1.6465	0.11	3
4	4.02	8.3815	0.08	1.3
5	4.15	4.8207	0.035	2.2

We cannot compare titratable acidity and total alkalinity factor between each group without accounting for variation in total solids used in each recording event. Therefore, titratable acidity and total alkalinity would be divided by total solids by each event to form x and y parameters respectively. These parameters are merely to allow for objective comparison between samples. This project will label parameters gotten from the control sample as x1 and y1 and that gotten from the stored samples as x2 and y2.

Table 4.3 below shows the x and y parameters for the control samples.

Control measurements			
Day	X1	Y1	Total solids (%)
0	0.53	0.28	5.3
1	0.59	0.39	5.1
2	0.18	0.12	6.9
3	0.19	0.04	3.4
4	3.67	0.05	1.7
5	2.56	0.03	1.8

Table 4.4 below shows the x and y parameters for the stored samples.

Stored item measurements			
Day	X2	Y2	Total solids (%)
0	0.53	0.28	5.3
1	0.65	0.21	3.9
2	0.75	0.19	3.2
3	0.55	0.04	3.0
4	6.45	0.06	1.3
5	2.19	0.02	2.2

The chart below shows the variation between the x parameters of both samples.

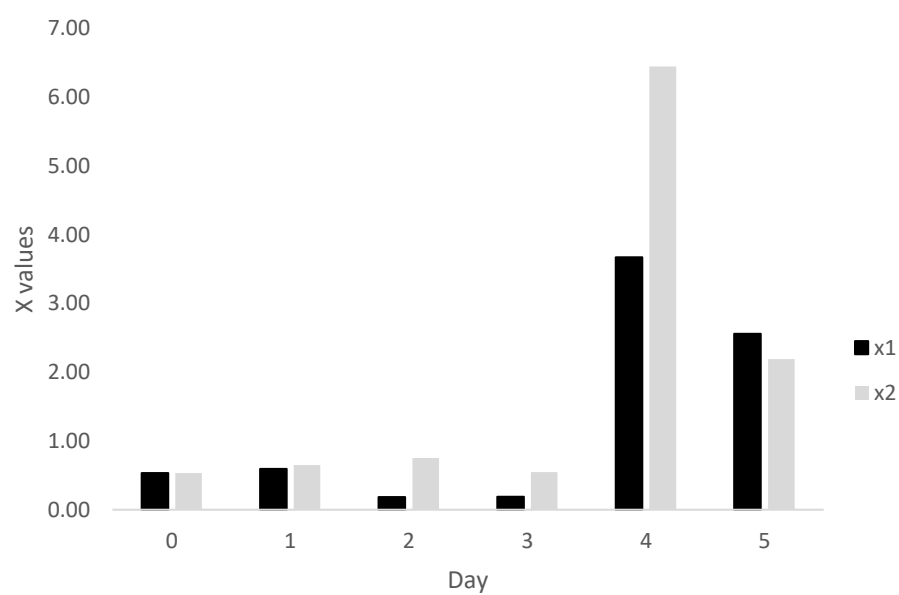


Figure 4.1: Bar chart for the X parameter of both Samples

The chart below shows the variation between the y parameters of both samples.

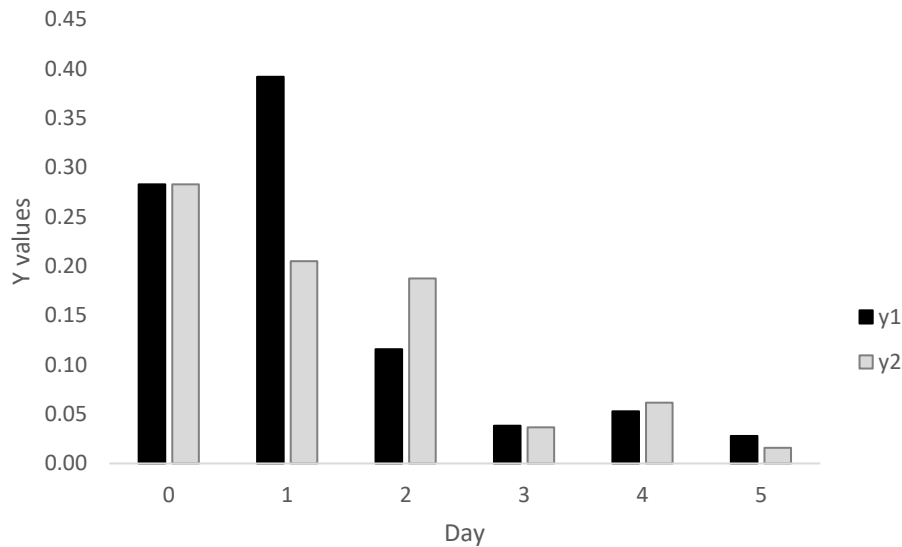


Figure 4.2: Bar chart for the Y parameter of both Samples

4.2 Discussion

The initial observation time of tomato stipulated in the procedure was seven days. In practice however, the obtained tomato samples were observed to have spoilt beyond human consumption by day 5. Signs and symptoms observed to reach such conclusions included extreme water loss, fragility of tomato skin during fluid extraction, unwanted smells, and the presence of white growths on some tomato samples further discussed in this chapter.

Literature agrees that all acid related parameters experience decrease as spoilage occurs. Titratable acidity decreases as spoilage occurs (Genanew, 2013). This is observed initially in both samples gotten although there seems to be an uncharacteristic increase in titratable acidity values for both samples at day 4 and 5. The range of titratable acidity also seems to align partially with the observed range of 0.891% to 0.25% with values as low as 0.165% observed during experimentation (Tigist et al., 2013). Total alkalinity projects a more coherent picture, decreasing steadily throughout the experiment on both samples. Total solids is less certain and

is not particularly effective in making conclusions by itself, usually used to agree with other parameters.

The PH values recorded in this experiment also somewhat falls in the expected range recorded in literature. Wade and Beuchat (2003) recorded a PH range of 4.3 to 5.8 for sound tomato pericarp tissues and 4.7 to 7.8 for decaying tomato pericarp tissues. In the result gotten from this experiment, PH values as low as 3.91 were recorded deviating from normal pre-established tomato PH ranges. This doesn't mean it is wrong as PH values of 3.90 was also observed by Asplund and Nurmi (1991).

Looking at the data, it cannot be inferred that any of the samples either stored under the modified atmosphere of Carbon dioxide or in air did better or worse than the other as every parameter measured followed the same trend without important differences between them. X1 and X2 parameters representing titratable acidity was tested for statistical significance giving rise to a p-value of 0.3131. The Y1 and Y2 parameters representing total alkalinity was tested for statistical significance giving a p-value of 0.4061, both being much higher than the universally accepted threshold of 0.05. Finally, the two groups of PH values were tested for statistical significance with a p-value of 0.4505 being recorded. This indicates that there is no statistically significant difference between the two methods of storage applied in this project.

CHAPTER 5

5.0 CONCLUSION AND RECOMMENDATION

5.1 Conclusion

In conclusion, there seems to be negligible difference in the quality of tomato plants stored in the improvised modified atmosphere of Carbon dioxide compared to that stored in air. The results of the experiment may have been because of the unanticipated growth of anaerobic micro-organisms on the tomato. This is backed up by the observance of white filamentous growths on some tomato samples under the modified atmospheric storage method. Another reason may be because the temperature at which the experimentation took place. As stated in the procedure, all process except specifically stated occurred at room temperature. Perhaps, the temperature makes it easy for respiration, enzymatic reactions, and microbial growth to occur even under anaerobic conditions. The conclusion remains the same: there seems to be no appreciable differences in product quality of tomato stored in air or in a carbon dioxide rich atmosphere under room temperature.

5.2 Recommendations

- This study should be repeated at low temperatures to determine if the difference in shelf-life of both samples can be compounded by the improvised modified atmospheric storage method.
- A follow up study should be carried out to identify the microorganisms observed and elucidate whether anaerobic atmosphere created by the dry ice could have favoured the growth.